AP Precalculus: 2026–2027

Unit 4: Functions Involving Parameters, Vectors, and Matrices Lesson 4.7: Parametrization of Implicitly Defined Functions (2 instructional periods, 55 minutes each)

Primary Objective: Students will express implicitly defined curves — ellipses, circles, and parabolas — as parametric functions using trigonometric parametrizations and algebraic substitutions, and apply parametric analysis (extrema from component functions, axis intercepts from zeros) to these curves. *(Big Idea: Functions)*

Priority Ideas & Skills

AP Skills	Key Understandings (EKs)
1.B *Express in equivalent forms* — Rewrite an implicitly defined equation (circle, ellipse, parabola) as a pair of parametric component functions that trace the curve, choosing a parameter substitution that makes the implicit equation a Pythagorean identity or a direct algebraic substitution.	**EK 4.7.A.1**: An implicitly defined equation in two variables can often be expressed as a parametric function by introducing a parameter t that traces the curve.
2.A *Identify information from representations* — Given a parametric pair $(x(t), y(t))$ derived from an implicit curve, extract the extrema of x and y, the axis intercepts, and the direction of motion as t increases.	**EK 4.7.A.2**: For an ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, the parametrization $x(t) = h + a\cos t$, $y(t) = k + b\sin t$, $0 \le t \le 2\pi$, traces the ellipse counterclockwise.
	EK 4.7.A.3: For a parabola $y - k = a(x-h)^2$, one parametrization is $x(t) = h + t$, $y(t) = k + at^2$ for $t \in \mathbb{R}$ (or a restricted domain). **EK 4.7.A.4**: Once parametrized, parametric analysis tools (direction of motion, extrema, intercepts) can be applied to curves that are not functions in the Cartesian sense.

Vocabulary, Concepts, & Theorems

Term	Definition / Note
implicit curve	A curve defined by an equation relating x and y that cannot always be written as a single explicit function $y = f(x)$; e.g. $\frac{x^2}{9} + \frac{y^2}{4} = 1$ (ellipse).
parametrization	A pair of component functions $(x(t), y(t))$ that traces an implicit curve as t varies; the parameter t may represent time or another quantity.
ellipse parametrization	$x(t) = h + a\cos t,\ y(t) = k + b\sin t,\ 0 \le t \le 2\pi;$ derived from the Pythagorean identity $\cos^2 t + \sin^2 t = 1$.
parabola parametrization	For $y - k = a(x-h)^2$: set $x(t) = h + t$, $y(t) = k + at^2$, $t \in \mathbb{R}$; the vertex occurs at $t = 0$, giving (h, k).

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8 minutes):

- **Unit circle parametrization** (Lesson 4.4): $(x(t), y(t)) = (\cos t, \sin t)$ for $0 \le t \le 2\pi$ — the foundation for all trig-based parametrizations.
- **Ellipse intercepts from standard form** (Lesson 4.6): reading a and b from $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to locate x- and y-intercepts.
- **Evaluating parametric component functions at specific** t (Lesson 4.2): computing $(x(t), y(t))$ at given values and checking whether those points satisfy an implicit equation.

Hook — Entry Event

Scenario: The ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ is *not* a function — it fails the vertical line test. But what if a particle traces this ellipse over time? Then we can ask: *Where is the particle at $t = \pi/3$? When does it cross the y-axis? What is its leftmost position?* These are parametric questions, and answering them requires converting the implicit equation into a parametric form.

Discussion prompts:

1. *"The identity $\cos^2 t + \sin^2 t = 1$ looks a lot like $\frac{x^2}{9} + \frac{y^2}{4} = 1$ if you write $x/3 = \cos t$ and $y/2 = \sin t$. How could you use that observation?"*
2. *"Once you have $(x(t), y(t))$, how do you find when the particle reaches its rightmost point?"*

Key tension: Implicit equations describe the *shape*; parametric forms add *motion* — direction, position at time t, and extrema using tools already developed in Lessons 4.2 and 4.3.

Lesson — Instruction Sequence (Period 1)

Step 1 — Review: Unit Circle and Scaling (6 min)
Revisit $(x(t), y(t)) = (\cos t, \sin t)$, $0 \le t \le 2\pi$. Key identity: $\cos^2 t + \sin^2 t = 1$. Scaling: if $x = a\cos t$ and $y = b\sin t$, substitute: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2 t + \sin^2 t = 1$. Ask: *"When $a \ne b$, what shape do we trace?"* An ellipse. This is the core idea of EK 4.7.A.1.

Step 2 — Ellipse Parametrization (15 min)
Standard form: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$. Let $\frac{x-h}{a} = \cos t$ and $\frac{y-k}{b} = \sin t$, so:

$$x(t) = h + a\cos t, \quad y(t) = k + b\sin t, \quad 0 \le t \le 2\pi.$$

Verify: $\frac{(h+a\cos t-h)^2}{a^2} + \frac{(k+b\sin t-k)^2}{b^2} = \cos^2 t + \sin^2 t = 1$. ✓ At $t = 0$: $(h + a,\ k)$ — rightmost. At $t = \pi/2$: $(h,\ k + b)$ — top.
Work through: $\frac{x^2}{16} + \frac{y^2}{4} = 1$. $h = 0$, $k = 0$, $a = 4$, $b = 2$. Parametrization: $x(t) = 4\cos t$, $y(t) = 2\sin t$, $0 \le t \le 2\pi$. (EK 4.7.A.2)

Step 3 — Parabola Parametrization (12 min)
For $y - k = a(x - h)^2$: x is unrestricted, so set $x(t) = h + t$, giving $y(t) = k + at^2$, $t \in \mathbb{R}$. Vertex at $t = 0$: (h, k). For a sideways parabola $x-h = a(y-k)^2$: set $y(t) = k + t$, $x(t) = h + at^2$.
Example: $y = x^2-4x+3 = (x-2)^2-1$. Vertex $(2, -1)$, $a = 1$. Parametrization: $x(t) = 2 + t$, $y(t) = -1 + t^2$. Evaluate at $t = -2, -1, 0, 1, 2$: $x = 0, 1, 2, 3, 4$ and $y = 3, 0, -1, 0, 3$. (EK 4.7.A.3)

Step 4 — Apply Parametric Analysis (10 min)
For the ellipse $x(t) = 3\cos t$, $y(t) = 5\sin t$:

- Rightmost point: $x(t) = 3$ when $t = 0 \Rightarrow (3, 0)$.
- Leftmost point: $x(t) = -3$ when $t = \pi \Rightarrow (-3, 0)$.
- y-axis crossing ($x(t) = 0$): $\cos t = 0$, so $t = \pi/2,\ 3\pi/2$. Points: $(0, 5)$ and $(0, -5)$.
- x-axis crossing ($y(t) = 0$): $\sin t = 0$, so $t = 0,\ \pi$. Points: $(3, 0)$ and $(-3, 0)$.

Reinforce: EK 4.7.A.4 — parametric tools apply to non-function curves.

Pacing note: Steps 1–2 fill Period 1; Steps 3–4 open Period 2 before activity.

Active Monitoring

- **Mixing up a and b**: Students may put b under x and a under y. Prompt: *"Which denominator appears below $(x-h)^2$ in standard form? That value is a^2, so a goes with x."*
- **Forgetting the center shift**: Students write $x(t) = a\cos t$ even when $h \neq 0$. Ask: *"At $t = 0$, your formula gives $x = a$. What should the rightmost point of this ellipse be?"*
- **Domain for parabola**: Students may restrict t to $[0, 2\pi]$ by analogy with ellipses. Clarify: for parabolas, t runs over all reals (the curve extends without bound).
- **Extrema by picture vs. component function**: Students read extrema from the Cartesian graph rather than solving the component equation. Reinforce: set $x(t)$ equal to its maximum value and solve for t; same for $y(t)$.

Group Work & Differentiation

Students work in groups of 3–4 on the Mission Control activity (tiered).

- **Tier R (Remediate)**: Given the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, verify that $x(t) = 4\cos t$, $y(t) = 3\sin t$ satisfies the equation at $t = 0$, $t = \pi/2$, and $t = \pi$.
- **Tier A (Approaching Proficiency)**: Write a parametrization for $\frac{(x-2)^2}{25} + \frac{(y+1)^2}{9} = 1$ and find the rightmost and leftmost extreme positions using the parametrization.
- **Tier E (Extension)**: For the sideways parabola $x - 1 = 2(y+3)^2$, write a parametrization with $y(t) = -3 + t$, identify the vertex coordinates, and find the axis intercepts by solving $x(t) = 0$ and $y(t) = 0$.

Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 2).

1. Write a parametrization for $\dfrac{(x+1)^2}{9} + \dfrac{(y-2)^2}{16} = 1$.
2. Using your parametrization, find the coordinates of the topmost and bottommost points on the ellipse.

Data use: Students who cannot perform the trig substitution without prompting need a reference card linking $\cos^2 t + \sin^2 t = 1$ to ellipse standard form; revisit at the start of the next lesson. Students who find extrema by returning to the Cartesian picture rather than solving the component equation need additional parametric analysis practice (Lesson 4.2 review).

Reinforcement & Extension

Homework (Lesson 4.7): 5–6 problems covering: parametrizing ellipses and circles in standard form, parametrizing parabolas by substitution, finding extrema and intercepts from the parametric form, and one AP-style multiple-choice item comparing two valid parametrizations of the same ellipse.
Extension: For the ellipse $x(t) = a\cos t$, $y(t) = b\sin t$, what substitution traces the ellipse *clockwise*? How does replacing t with $-t$ affect $\cos t$ and $\sin t$ separately?
Preview of Lesson 4.8: Ask students: *"From $x = h + t$, $y = k + at^2$, can you eliminate t to recover a Cartesian equation? What equation in x and y do you get?"* This reversal — parametric back to implicit — will be the focus next.